If we returned the tasks in the same order as we insert them, we would have just implemented as a plain queue (see figure 2.2 for a quick glance at how a queue works, and appendix C for a description of basic containers). Instead, let’s assume that now we have our priority queue containing those five elements: we still don’t know the internals of the PQ but we can only query it through its API.

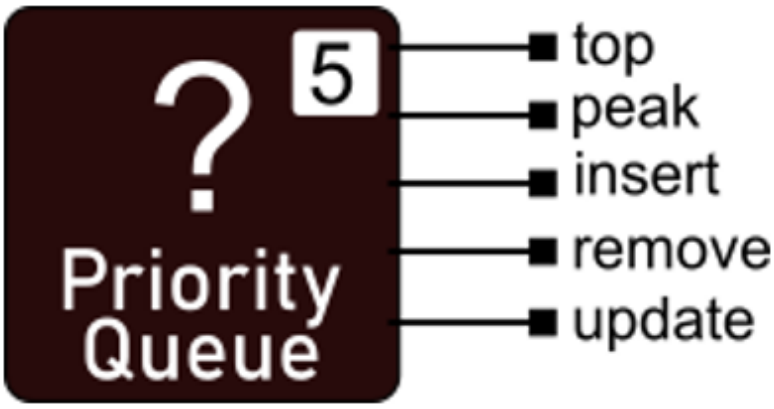


Figure 2.1 Representation of a priority queue as a black box. If we employ an implementation of a priority queue provided by a third-party library (or from standard libraries), and we trust this implementation to be correct, we can use it as a black box. In other words, we can ignore its internals, and just interact with it through its API.

For instance, we can check how many elements it contains, and even take a peek at the one at the top (figure 2.1).

Or we can directly ask it to return us the top element (the one with the highest priority) and remove it from the queue.